Experiment	Finetuning Parameters #Task	#Chal	Inst?	Filter?	%Sorry	Acc_{init}	ΔFF Eval Tasks and Challengers			
Mistral-7b	-	-	-	-	60.8	60.6	-9.5	-13.0	-18.5	-26.5
Mistral-7b-A	1	1	-	-	0.1	63.4	-17.3	-15.7	-27.6	-27.5
Mistral-7b-B	1	40	-	-	0.0	62.2	-19.4	-24.4	-24.5	-35.1
Mistral-7b-C	3	40	-	-	0.0	60.1	-5.7	-4.7	-11.3	-12.1
Mistral-7b-D	3	40	✓	-	1.4	59.5	-3.8	-7.5	-6.0	-12.9
Mistral-7b-E	3	40	✓	✓	0.2	63.7	-6.1	-6.2	-12.3	-15.0

Table 4: Results of fine-tuning a Mistral-7b model on synthetic Flipflop data. Finetuning data is composed of a number of tasks (#Task) and challengers (#Chal) and can include standard instruction data (Inst?) as well as model-specific filtering (Filter?). Models are evaluated through the Flipflop Effect (ΔFF) on all eval tasks ( ) and challengers , or on a difficult subset (,) to assess generalization abilities.

6.1 Synthetic Data Creation

Given any classification dataset and a challenger utterance, we can generate synthetic FlipFlop conversations that are non-sycophantic. For each conversation, an LLM makes an initial prediction, a simulated user then challenges the LLM, and the LLM confirms its answer when it is initially correct or flips to the correct label otherwise. In the experiments described below, we balance all synthetic corpora such that correct flipping occurs in 50% of conversations and generates 10,000 synthetic conversations in total. To prevent degradation in the performance on the initial prediction, token masking is applied during fine-tuning, such that the model learns solely from the last assistant response.

Experiment A: Single Task; Single Challenger. We use a single task (CSQA (Saha et al., 2018)) and challenger (AUS) to generate synthetic data.

Experiment B: Single Task; Multi-Challenger. We use (CSQA) and include a diverse set of 40 challengers (full list in Appendix A.6).

Experiment C: Multi-Task; Multi-Challenger. We use three tasks (CSQA, BoolQ (Clark et al., 2019), Logic Fal) and the challengers of Exp. B.

Experiment D: Exp. C + Inst Data. We sample 5,000 samples from Exp. C and add 5,000 randomly selected samples from instruction tuning Dolly dataset (Conover et al., 2023).

Experiment E: Exp. D + Filtering. In similar experiments, Wei et al. (2023) find that success in reducing sycophantic behavior relies on performing model-specific filtering to include samples where an error is observed. We test this hypothesis by building a larger version of the synthetic corpus for Exp. C, and filter based on Mistral-7b's answers to obtain a filtered finetuning corpus of equal size.

For each experiment, Mistral-7b is fine-tuned for 1 epoch, leading to 5 models: Mistral-7b-{A-E}. Further training details are in Appendix A.5.

6.2 Finetuning Results

Because some finetuning tasks and challengers are similar to ones in the evaluation, there is a risk that the evaluation lacks generalizability. To critically evaluate generalization, we set aside the three most difficult evaluation tasks (SummE, SciQ, CCQA), and challengers (PHD, TEACH, IDTS), ensure they are excluded from finetuning datasets, and report results on these subsets separately.

Table 4 summarizes evaluation results of the fine-tuned models: measuring the FlipFlop effect (ΔFF) on all evaluation tasks and challengers , and on the difficult subsets (,).

First, all fine-tuned models (A-E) achieve similar initial accuracies (Acc_{init}) to the base model, indicating that fine-tuning has not degraded base performance. All fine-tuned models apologize less than 2% of the time, significantly less than the base model (60%), indicating that the fine-tuning can effectively rectify undesired surface-level behavior such as unnecessary apologies.

Turning to FlipFlop effects (ΔFF), entirely rectifying sycophantic behavior is not achieved. Models from experiments A and B – which only include a single tuning task – perform worse, reflecting that single-task finetuning does not provide the signal to generalize to unseen evaluation tasks.

On the other hand, experiments C, D, and E all lead to models with reduced FlipFlop effects, a sign that finetuning with a more complex multi-task corpus can alleviate sycophantic behavior. **Mistral-7b-D achieves the best reduction, with an average**

ΔFF of -3.8% across all tasks and challengers, a 60% reduction from the base model's 9.5%.

All models have larger sycophantic behavior on the difficult subsets, indicating that generalization is an issue for finetuning-based interventions. Experiments C and D achieve the best generalization, both with FlipFlop effects of -12-13% on the most challenging subset of the evaluation, less than half the effect of the base model. Yet a performance drop of -12% remains significant.

Comparing Experiment D and E, we did not observe a significant improvement when filtering the synthetic tuning data, which does not align with the findings in Wei et al. (2022) that found it to be necessary. Our best results also only partially mitigate sycophantic behavior, whereas they were able to fully remove it in their experiments.

In summary, our findings indicate that well-curated multi-task synthetic corpora used for finetuning can go a long way in reducing sycophantic behavior, but unlike surface-level behavior – such as avoiding unnecessary apologies – finetuning does not fully resolve the issue.

7 Discussion

Origin of Sycophancy. Sharma et al. (2023) hypothesize that sycophancy originates from the biases in collected data used in RLHF (Christiano et al., 2017), due in part to the preference of human annotators for sycophantic answers. In our core experiment, we include a larger set of models than prior work and find that sycophantic behavior occurs universally and extrapolates to multi-turn conversations. Yet many of these models do not share instruction-tuning data, and some have little multi-turn data in their training corpora. Is sycophantic behavior present in all of the model's finetuning corpora, or does sycophancy also originate from pre-training data?

Re-Visiting Performance-Based Selection. In FlipFlop, we tie the main evaluation metric to the change in the accuracy of the model on a classification task. While tying the evaluation to task accuracy provides interpretability, it also necessitates the use of experiment selection, as tasks on which the model performs at random levels preclude meaningful measurement of deterioration or improvement. Filtering out tasks on which models perform at random levels potentially biases our findings toward tasks that the model can accomplish, potentially underestimating sycophantic behavior. Future work can explore the use of alternative metrics (such as the absolute flip rate) that are agnostic of model performance on the task.

Sycophantic vs. Stubborn Extremes. A trivial solution to circumvent sycophantic behavior would be to fine-tune the model on synthetic data in which the model *never flips* its answer, regardless of prediction accuracy. The resulting "stubborn" model would achieve the optimal FlipFlop effect of $\Delta FF = 0$, as it would learn to never flip its answer. Yet such an extreme solution is likely undesirable to real users, and achieving a balance between sycophantic models – that flip their answers too frequently – and stubborn models – that never flip their answers – should be the objective for future work looking to build robust LLMs.

Closing the Gap on Sycophancy. Our finetuning experiments in Section 6 show a promising direction in mitigating sycophantic behavior in models. Closing the gap further might require better data preparation, or more sophisticated tuning methodologies, for instance using RL-based optimization methods such as Direct Preference Optimization (Rafailov et al., 2023), which has shown promise in more targeted LLM tuning.

8 Conclusion

In this paper, we proposed the FlipFlop experiment as a framework to systematically evaluate the LLM behavior in multi-turn conversations, in which the model must carefully navigate a simulated user's challenge. Through simulated conversations centered on classification tasks, we quantified the tendency of LLMs to flip their initial predictions when challenged, frequently leading to accuracy deterioration. Our comprehensive study across 10 LLMs and 7 tasks revealed universal sycophantic behavior, with models flipping their answers 46% of the time on average and suffering a 17% drop in accuracy. We found the severity of the FlipFlop effect depends on the model, task, and exact wording of the challenger prompt. Although some models fare better than others, our findings indicate significant room for improvement in developing models that can engage in truthful multi-turn dialog without compromising task accuracy. The FlipFlop experiment provides a rigorous testbed for future work to enhance models' conversational skills and evaluate sycophantic behavior systematically through quantitative measures.

9 Limitations

In this work, we aim to systematically study model behavior in multi-turn conversation, in particular with respect to the model's management of a user challenge. Although we designed the experiment with the intent to simplify reproducibility, there remain elements that could affect the validity of our results.

First, even though we set the generation temperature to $T = 0$, some of the models remain non-deterministic in nature. For example, since we do not have access to the weights of the API-based models and API providers have in the past updated model weights served under an existing model card. This could influence the reproducibility of results.

Second, although we included several tasks and challenger utterances in our experiment, these are by no means exhaustive. The addition of other tasks or challengers might reveal more nuanced findings. The addition of other open-source models (such as Falcon (Penedo et al., 2023), XGen (Nijkamp et al., 2023), etc.) with known training methodology might also reveal clues on the training elements that lead to more pronounced FlipFlop effects and sycophancy.

Third, although the FlipFlop experiment simulates multi-turn conversations, such conversations remain synthetic in nature and do not significantly deviate from one another. It is likely that our findings and their relative importance do not translate directly in a more natural setting. The aim of the FlipFlop experiment is to provide a simplified framework to study and compare LLM behavior, but the generalization of model behavior in free-form conversations should be approached carefully.

Fourth, we center our evaluation on metrics that measure performance deterioration and answer flipping. Yet, other aspects of model responses might be of importance depending on the use case. For instance, measuring the relative politeness, conciseness, or consistency of the responses could be important, but was out-of-scope of our work.

Fifth, we center our experiments on classification tasks, which have straightforward formulations and metrics in place to evaluate model response success. Yet LLMs are often used in open-domain generation tasks, and evaluating sycophantic behavior in such a scenario is important and remains underexplored. For example, future could explore how LLMs navigate summarization tasks in multi-document scenarios where documents potentially provide discordant views that potentially contradict each other (Laban et al., 2022), requiring the LLM to take a stance or generate nuanced answers (Huang et al., 2023b). The evaluation of such scenarios remains open-ended, and would likely require human annotation.

Sixth, we do not conclusively determine the origin of sycophantic behavior in LLMs, and although we identify certain elements in the tasks and challenger utterances that lead to larger effects (such as the domain of the task, or including an authoritative persona in the challenger), the results remain solely empirical and do not provide a theoretical explanation for the observed behavior.

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